

Albuterol (salbutamol): Drug information - UpToDate

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For additional information see "Albuterol (salbutamol): Patient drug information" and "Albuterol (salbutamol): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- ProAir Digihaler [DSC];
- ProAir HFA [DSC];
- ProAir RespiClick;
- Proventil HFA [DSC];
- Ventolin HFA

Brand Names: Canada

- Airomir;
- APO-Salbutamol HFA;
- DOM-Salbutamol;
- PMS-Salbutamol;
- TEVA-Salbutamol;
- TEVA-Salbutamol HFA;
- Ventolin;
- Ventolin Diskus;
- Ventolin HFA;
- Ventolin PF [DSC]

Pharmacologic Category

- Beta₂ Agonist

Dosing: Adult

Dosage guidance:

Dosage form information: Unless otherwise noted, dosing for the metered-dose inhaler (MDI) and dry powder inhaler (DPI) formulations is based on US products (90 mcg/actuation). A valved holding chamber (spacer) is recommended for use with an MDI. A DPI does not require use of a valved holding chamber.

Asthma

Asthma:

Acute exacerbation:

Mild to moderate exacerbations (initial home management): **Note:** Patients with worsening symptoms despite initial care should seek immediate medical attention (Ref).

Metered-dose inhaler or dry powder inhaler (90 mcg/actuation): **Oral inhalation:** 2 to 4 inhalations every 20 minutes for 3 doses; if good response, can lengthen interval to every 3 to 4 hours as needed; if incomplete response, can lengthen interval to every 1 to 3 hours as needed (Ref).

Nebulization solution: **Oral inhalation:** 2.5 mg every 20 minutes for 3 doses; if good response, can lengthen interval to every 3 to 4 hours as needed; if incomplete response, can lengthen interval to every 1 to 3 hours as needed (Ref).

Moderate to severe exacerbations (management in primary or acute care settings): **Note:** For severe exacerbations, albuterol is used in combination with an inhaled short-acting muscarinic antagonist. Nebulized therapy may be preferred in patients who have more severe symptoms or who cannot effectively use an inhaler (Ref).

Metered-dose inhaler or dry powder inhaler (90 mcg/actuation): **Oral inhalation:** 4 to 10 inhalations every 20 minutes for 3 doses, then taper as tolerated (eg, to 2 to 4 inhalations every 1 to 4 hours as needed) (Ref). High doses are typically administered in a monitored setting.

Nebulization solution: **Oral inhalation:** 2.5 to 5 mg every 20 minutes for 3 doses, then taper as tolerated (eg, to 2.5 to 5 mg every 1 to 4 hours as needed). For patients who are critically ill, up to 10 to 15 mg may be administered by continuous nebulization over 1 hour via special apparatus (Ref).

IV continuous infusion [Canadian product]: **Note:** Reserve intravenous beta-agonists for those patients in whom inhaled therapy cannot be used reliably (Ref). Initial: 5 mcg/minute; may increase up to 10 to 20 mcg/minute at 15- to 30-minute intervals, if needed.

Intermittent symptom relief (alternative agent): **Note:** Use in combination with an inhaled corticosteroid (preferred; may also be used alone) on an as-needed basis (reliever therapy) rather than regularly scheduled. For maintenance therapy, additional maintenance treatments should be used (Ref).

Nebulized therapy may be preferable for patients who have more severe symptoms or who cannot effectively use an inhaler (Ref).

Metered-dose inhaler or dry powder inhaler (90 mcg/actuation): **Oral inhalation:** 2 inhalations every 4 to 6 hours as needed (Ref); some experts recommend up to 4 inhalations every 4 to 6 hours for moderate to severe symptoms (Ref).

Canadian formulation: Ventolin Diskus [Canadian product]: DPI (200 mcg/inhalation): **Oral inhalation:** 1 inhalation every 4 to 6 hours as needed.

Nebulization solution: **Oral inhalation:** 2.5 mg every 4 to 6 hours as needed (Ref).

Maintenance therapy (alternative agent) (adjunctive therapy): **Note:** Oral formulations of albuterol (eg, tablet, syrup) are less effective than inhaled forms for bronchoconstriction and more likely to cause adrenergic adverse effects. For maintenance therapy, additional maintenance treatments (eg, inhaled corticosteroids) are recommended (Ref).

Syrup, tablets: **Oral: Initial:** 2 to 4 mg three to four times daily; if desired response not achieved, may gradually increase dose. Maximum daily dose: 32 mg/day.

Exercise-induced bronchoconstriction (prevention):

Metered-dose inhaler or dry powder inhaler (90 mcg/actuation): **Oral inhalation:** 2 inhalations 5 to 20 minutes prior to exercise (Ref).

Canadian formulation: Ventolin Diskus [Canadian product]: DPI (200 mcg/inhalation): **Oral inhalation:** 1 inhalation 15 minutes prior to exercise.

Bronchospasm due to anaphylaxis

Bronchospasm (moderate to severe) due to anaphylaxis (adjunct to epinephrine): Note: Administer epinephrine first when treating anaphylaxis; administer albuterol for residual respiratory symptoms not responding to epinephrine. Do **not** use albuterol for initial or sole treatment of anaphylaxis because albuterol does **not** prevent or relieve upper airway edema, hypotension, or shock (Ref).

Metered-dose inhaler or dry powder inhaler (90 mcg/actuation): **Oral inhalation:** 2 to 3 inhalations as needed for symptom relief (Ref). For patients requiring mechanical ventilation, up to 8 inhalations may be used, if needed (Ref).

Nebulization solution: **Oral inhalation:** 2.5 to 5 mg; repeat as needed (Ref).

Chronic obstructive pulmonary disease

Chronic obstructive pulmonary disease:

Acute exacerbation: Note: Although similar efficacy exists among formulations, some experts prefer nebulized therapy during severe chronic obstructive pulmonary disease exacerbations. Not recommended as monotherapy for hospitalized patients, unless there are clear side effects to antimuscarinic therapy or if additional bronchodilation is needed after ipratropium dosing limits are reached (Ref).

Metered-dose inhaler or dry powder inhaler (90 mcg/actuation): **Oral inhalation:** 2 inhalations every 1 to 2 hours (up to 3 doses), then every 2 to 4 hours, as needed; for patients requiring emergency department or hospital-based care, may increase to 4 inhalations every 1 to 2 hours (up to 3 doses). For patients requiring mechanical ventilation, up to 8 inhalations may be used, if needed (Ref).

Nebulization solution: **Oral inhalation:** 2.5 mg every 1 to 2 hours (up to 3 doses), then every 2 to 4 hours as needed (Ref).

Intermittent symptom relief : Note: Use for intermittent symptoms on an as-needed basis rather than regularly scheduled maintenance therapy. Typically used in combination with inhaled short-acting muscarinic antagonist; combination therapy provides greater improvement in FEV₁ and symptoms over monotherapy with a short-acting beta-2 agonist (Ref).

Metered-dose inhaler or dry powder inhaler (90 mcg/actuation): **Oral inhalation:** 2 inhalations every 4 to 6 hours as needed (Ref).

Canadian formulation: Ventolin Diskus [Canadian product]: Dry powder inhaler (200 mcg/inhalation): **Oral inhalation:** 1 inhalation every 4 to 6 hours as needed.

Nebulization solution: **Oral inhalation:** 2.5 mg every 4 to 6 hours as needed (Ref).

Hyperkalemia

Hyperkalemia (alternative agent) (adjunct) (off-label use): Note: May use after administration of standard therapies (calcium, insulin with dextrose, and potassium removal therapy) for those with continued symptoms or serious ECG manifestations or in patients for whom dialysis is not appropriate or feasible (Ref).

Oral inhalation: Nebulization solution: 10 to 20 mg via nebulization over 10 minutes (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function:

Oral inhalation: No dosage adjustment necessary (relatively low systemic absorption) (Ref).

IV continuous infusion [Canadian product]:

eGFR >30 mL/minute/1.73 m²: No dosage adjustment necessary.

eGFR <30 mL/minute/1.73 m²: A substantial amount is eliminated in the urine (~65%), and albuterol may accumulate (Ref). No initial dosage adjustment necessary but use with caution and monitor for adverse effects. Where feasible, switch to inhaled formulations. Reduced doses may be necessary if used for more than 24 hours (Ref).

Hemodialysis, intermittent (thrice weekly):

Oral inhalation: Unlikely to be dialyzed (large V_d): No supplemental dose or dosage adjustment necessary (Ref).

IV continuous infusion [Canadian product]: Unlikely to be dialyzed (large V_d): Dose as for eGFR <30 mL/minute/1.73 m² (Ref).

Peritoneal dialysis:

Oral inhalation: Unlikely to be dialyzed (large V_d): No dosage adjustment necessary (Ref).

IV continuous infusion [Canadian product]: Unlikely to be dialyzed (large V_d): Dose as for eGFR <30 mL/minute/1.73 m² (Ref).

CRRT:

Oral inhalation: No dosage adjustment necessary (Ref).

IV continuous infusion [Canadian product]: Dose as for eGFR <30 mL/minute/1.73 m² (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration):

Oral inhalation: No dosage adjustment necessary (Ref).

IV continuous infusion [Canadian product]: Dose as for eGFR <30 mL/minute/1.73 m² (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Older Adult

Oral inhalation: Refer to adult dosing.

Oral (syrup, tablets): Initial: 2 mg 3 to 4 times daily initially; if control not achieved, may gradually increase dose up to a maximum of 32 mg/day.

Dosing: Pediatric

(For additional information see "Albuterol (salbutamol): Pediatric drug information")

Dosage guidance:

Clinical considerations: A valved holding chamber or spacer is recommended for use with a metered-dose inhaler (MDI) in patients with poor technique and is the recommended medication delivery system in all patients ≤5 years of age, with the addition of a mask for patients <3 or 4 years of age or until proper technique is demonstrated (Ref). A dry powder inhaler (DPI) does not require use of a valved holding chamber. Young children and patients with severe symptoms may have reduced inspiratory flow which may reduce drug deposition in the lung from a DPI (Ref).

Asthma

Asthma:

Note: Scheduled daily inhaled albuterol use is not recommended for long-term maintenance treatment (Ref). Increasing use or regular use >2 days/week for symptom control (except prevention of exercise-induced bronchospasm) indicates inadequate control and need for additional long-term control therapy (Ref).

Acute symptom relief: Note: For use as an as-needed reliever therapy for acute symptoms outside of an exacerbation. Beta agonists should be used in combination with inhaled corticosteroids (ICS);

in patients <4 years, use short-course daily ICS (at onset of viral illness) with as-needed albuterol; for patients ≥4 years of age, use as-needed albuterol immediately followed by ICS or a combination ICS and formoterol inhaler (Ref).

Metered-dose inhaler (MDI), dry powder inhaler (DPI): 90 mcg/actuation:

Infants, Children, and Adolescents: Limited data available in ages <4 years: Oral inhalation: 2 inhalations every 4 to 6 hours as needed (Ref); 1 inhalation every 4 hours may be sufficient in some patients (Ref).

Nebulization:

Infants and Children ≤4 years: Limited data available in ages <2 years: Oral inhalation: 0.63 to 2.5 mg every 4 to 6 hours as needed (Ref).

Children >4 to <12 years: Oral inhalation: 1.25 to 5 mg every 4 to 8 hours as needed (Ref).

Children ≥12 years and Adolescents: Oral inhalation: Usual dose: 2.5 mg every 6 to 8 hours as needed (Ref). Reported range: 1.25 to 5 mg every 4 to 8 hours as needed (Ref).

Acute exacerbation: Note: Use for progressively worsening asthma symptoms not responding to as-needed reliever therapy. Beta-agonists are often used in conjunction with systemic glucocorticoids in patients with acute exacerbations.

Home management: Note: Patients with severe symptoms or worsening symptoms despite initial care should seek immediate medical attention.

MDI, DPI: 90 mcg/actuation:

Infants: Oral inhalation: 2 inhalations every 20 minutes as needed for 2 to 3 doses; patients with more severe symptoms may require up to 4 inhalations/dose (Ref); if initial response is good, lengthen dosing frequency and administer 2 to 4 inhalations every 3 to 4 hours for 24 to 48 hours (Ref).

Children and Adolescents: Oral inhalation: 2 to 4 inhalations every 20 minutes as needed for 2 to 3 doses; patients with more severe symptoms may require up to 6 inhalations/dose (Ref); if initial response is good, lengthen dosing frequency and administer 2 to 6 inhalations every 3 to 4 hours for 24 to 48 hours (Ref).

Nebulization:

Infants and Children ≤4 years: Limited data available in ages <2 years: Oral inhalation: Usual dose: 1.25 to 2.5 mg every 20 minutes for 3 doses if needed; if initial response is good, lengthen dosing frequency and administer 1.25 to 2.5 mg every 3 to 4 hours for 24 to 48 hours; reported range: 0.63 to 2.5 mg/dose (Ref).

Children >4 to <12 years: Oral inhalation: Usual dose: 2.5 to 5 mg every 20 minutes for 3 doses if needed; if initial response is good, lengthen dosing frequency and administer 2.5 to 5 mg every 3 to 4 hours for 24 to 48 hours; reported range: 1.25 to 5 mg/dose (Ref).

Children ≥12 years and Adolescents: Oral inhalation: Usual dose: 2.5 mg every 20 minutes for the first hour if needed; if initial response is good, lengthen dosing frequency and administer 2.5 mg every 3 to 4 hours for 24 to 48 hours; reported range: 1.25 to 5 mg/dose (Ref).

Primary care/acute care management:

MDI, DPI: 90 mcg/actuation:

Infants and Children <12 years: Limited data available in ages <4 years: Oral inhalation: 4 to 8 inhalations every 20 minutes for 3 doses (Ref); for patients ≥6 years of age, some experts recommend doses up to 10 inhalations/dose (Ref). Weight-band dosing may also be used by some centers; doses based on patient weight as follows: 5 to <10 kg: 4 inhalations; 10 to <30 kg: 6 inhalations; ≥30 kg: 8 inhalations; doses are administered every 20 minutes for 3 doses (Ref). After the initial hour, dose may vary from 4 to 10 inhalations every 1 to 4 hours based on patient response (Ref).

Children ≥ 12 years and Adolescents: Oral inhalation: 4 to 8 inhalations every 20 minutes for 3 doses (Ref); some experts recommend doses up to 10 inhalations/dose (Ref). After the initial hour, dose may vary from 4 to 10 inhalations every 1 to 4 hours based on patient response (Ref).

Nebulization: **Note:** Nebulization is preferred for patients who are unable to effectively use an MDI or DPI due to age, agitation, or severity of the exacerbation (Ref).

Intermittent: Infants, Children, and Adolescents: Limited data available in ages < 2 years: Oral inhalation: 0.15 mg/kg/dose (minimum dose: 2.5 mg/dose; maximum dose: 5 mg/dose) every 20 minutes for 3 doses then 0.15 to 0.3 mg/kg/dose not to exceed 10 mg/dose every 1 to 4 hours as needed (Ref). **Note:** For severe exacerbations, addition of ipratropium may be considered if poor response to initial aggressive short-acting beta₂-agonist (SABA) therapy in an acute care setting (ie, emergency department). Ipratropium has not been shown to provide further benefit (eg, after first 24 hours) once the patient is hospitalized (Ref).

Continuous nebulization: Dosing regimens variable; optimal dosage not established; consult institutional-specific protocols.

Infants and Children < 12 years: Initial: 0.5 mg/kg/hour (Ref); in patients with severe exacerbations doses as high as 1 mg/kg/hour may be needed (Ref). Weight-band dosing has also been reported; doses based on patient weight have been used as follows: 5 to < 10 kg: 5 to 7.5 mg/hour; 10 to < 20 kg: 10 mg/hour; ≥ 20 kg: 15 to 20 mg/hour. Adjust dose as needed based on patient response; maximum dose: 20 mg/hour (Ref).

Children ≥ 12 years and Adolescents: 10 to 15 mg/hour (Ref); higher doses of up to 20 mg/hour have also been reported (Ref).

Bronchospasm, acute

Bronchospasm, acute:

Note: Use for acute bronchospasm outside of asthma.

Prevention: Limited data available; optimal dose not established. **Note:** Administer prior to anticipated exposure to triggers (eg, bronchospastic inhaled therapies), airway clearance techniques (eg, cystic fibrosis) or in high-risk patients (eg, history of asthma, wheezing, recent upper respiratory tract infection) prior to procedures (Ref).

Infants, Children, and Adolescents:

MDI, DPI: 90 mcg/actuation: Oral inhalation: 2 to 4 inhalations administered 10 to 30 minutes prior to trigger exposure, airway clearance techniques, or procedure.

Nebulization: Oral inhalation: 2.5 mg administered 10 to 30 minutes prior to trigger exposure, airway clearance techniques, or procedure; some patients may require doses up to 5 mg based on history and individual risk factors.

Treatment: Limited data available: **Note:** Albuterol has been used in the treatment of bronchospasm/wheezing due to conditions such as viral illness, contrast reactions, or as adjunct to epinephrine in the treatment of anaphylaxis (Ref).

Infants, Children, and Adolescents:

MDI, DPI: 90 mcg/actuation: Oral inhalation: 2 inhalations; repeat as needed; frequency may vary and range from every 4 to 6 hours as needed up to every 20 minutes for 3 doses based on patient response; patients with history of asthma may require higher doses (ie, 5 to 10 inhalations every 20 minutes) in some situations (eg, when used as adjunct therapy in anaphylaxis).

Nebulization: Oral inhalation: 2.5 to 5 mg; repeat as needed based on clinical condition, symptoms, and response; may give as frequently as every 20 minutes for up to 3 doses or may administer continuously if needed.

Exercise-induced bronchoconstriction, prevention

Exercise-induced bronchoconstriction, prevention:

Note: Tolerance to the protective effects of albuterol against exercise-induced bronchospasm may develop with regular daily use (Ref).

MDI, DPI: 90 mcg/actuation:

Children <4 years: Limited data available: Oral inhalation: 1 to 2 inhalations 5 to 20 minutes before exercising (Ref). **Note:** Pretreatment in patients <4 years of age should only be used when a specific triggering activity can be clearly identified.

Children ≥4 years and Adolescents: Oral inhalation: 2 inhalations 5 to 20 minutes before exercising (Ref); doses up to 4 inhalations have been reported (Ref).

Canadian formulation: DPI: 200 mcg/inhalation: Children ≥4 years and Adolescents: Oral inhalation: 1 inhalation 15 minutes before exercise.

Hyperkalemia

Hyperkalemia (adjunct therapy): Limited data available:

Note: Albuterol is administered to shift potassium intracellularly and may be administered concomitantly with IV insulin and dextrose infusion for additive effect (Ref). Albuterol should not be used as the sole agent for treating moderate to severe hyperkalemia; use in combination with therapies to stabilize myocardium and eliminate potassium (Ref). Use caution in patients with preexisting cardiac disorders (eg, arrhythmias) (Ref); patients should be on a cardiac monitor during administration.

Infants, Children, and Adolescents (Ref):

<25 kg: Nebulization: Oral inhalation: 2.5 mg nebulized over 10 minutes; dose may be repeated as needed.

25 to 50 kg: Nebulization: Oral inhalation: 5 mg nebulized over 10 minutes; dose may be repeated as needed.

>50 kg: Nebulization: Oral inhalation: 10 mg nebulized over 10 minutes; dose may be repeated as needed. **Note:** Doses as high as 20 mg have been suggested for use in adults and may be utilized for larger children and adolescents (Ref).

Bronchospasm, patients unable to use or tolerate inhaled therapy

Bronchospasm (long-term prevention), patients unable to use or tolerate inhaled therapy:

Note: Oral formulations of albuterol (eg, tablet, syrup) are not recommended; use is very rare because they are less effective than inhaled forms and more likely to cause adverse effects (Ref).

Children 2 to 6 years: Oral: Immediate release: Syrup, tablet: Initial: 0.1 mg/**kg**/dose (maximum dose: 2 mg/dose) 3 times daily; if desired response not achieved, gradually increase dose to 0.2 mg/**kg**/dose (maximum dose: 4 mg/dose) 3 times daily; maximum daily dose: 12 mg/**day** (Ref).

Children >6 and Adolescents ≤14 years: Oral: Immediate release: Syrup, tablet: Initial: 2 mg 3 to 4 times daily; if desired response not achieved, gradually increase dose; maximum daily dose: 24 mg/**day** (Ref).

Adolescents >14 years: Oral: Immediate release: Syrup, tablet: Initial: 2 to 4 mg 3 to 4 times daily; if desired response not achieved, gradually increase dose; maximum daily dose: 32 mg/**day** (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution when using high doses in patients with renal impairment. No dosage adjustment required, including patients on hemodialysis, peritoneal dialysis, or CRRT (Ref).

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in manufacturer's labeling.

Adverse Reactions (Significant): Considerations

Cardiovascular effects

Short-acting beta-2 agonists (SABAs), such as albuterol, may cause mild **tachycardia**, **cardiac arrhythmias** (less common), QTc interval changes, and more severe cases of cardiac ischemia, heart failure, and **cardiomyopathy** (Ref). Swallowing-induced atrial tachyarrhythmia (SIAT) (**atrial tachycardia**) has rarely been reported (Ref). **Supraventricular tachycardia** has been reported in a neonate (Ref). In pediatric patients receiving continuous albuterol, elevated troponin-T levels have been reported, the significance of which is unknown (Ref). Stress-induced cardiomyopathy (takotsubo cardiomyopathy) has also been reported (Ref).

Mechanism: Dose-related; may be related to the pharmacologic action; beta-2 agonist effects on the heart as well as loss of beta-2 agonist selectivity at higher doses (ie, beta-1 agonist effects) (Ref). Cardiac toxicity may occur in the setting of hypoxemia, secondary to vasodilation and possible pulmonary shunting, resulting in hypotension (Ref). SIAT may be linked to triggering of adrenergic reflexes in the esophagus (Ref). Stress-induced cardiomyopathy may be due to catecholamine-induced reduction in blood flow, resulting in obstruction and secondary ischemia of the left ventricle and anterior wall (Ref).

Onset: Rapid; tachycardia may occur within 15 minutes after administration (Ref). Cardiomyopathy may occur within days of administration (Ref).

Risk factors:

- Overuse (dose and frequency) (Ref)
- Concurrent use of other sympathomimetics (Ref)
- Preexisting cardiovascular disease or predisposition to cardiovascular effects (Ref)

CNS effects

Albuterol may cause CNS effects, including (but not limited to) **excitement**, **nervousness**, **tremor**, **anxiety**, **hyperactive behavior**, and **insomnia**. Excitement, nervousness, and insomnia may occur more frequently in younger pediatric patients (2 to 6 years) versus older pediatric patients and adults (Ref). There are also isolated case reports of **psychosis** (Ref).

Mechanism: Dose-related; related to the pharmacologic effect (ie, adrenergic CNS stimulation) (Ref).

Onset: Rapid; psychosis occurred within 2 days of overuse in one case report (Ref).

Risk factors:

- Overuse (dose and frequency) (Ref)

Paradoxical bronchospasm

Albuterol may cause **paradoxical bronchospasm** with an incidence in the literature ~4% (Ref).

Mechanism: Non-dose-related; may be related to a hypersensitivity reaction or irritation to excipients in albuterol inhalers triggering airway hyperresponsiveness (Ref).

Onset: Varied; occurred within 30 minutes in one case report (Ref); has also occurred after months of therapy (Ref).

Risk factors:

- Dosage form (metered-dose inhaler and nebulizer solution) (Ref)
- Benzalkonium chloride (Ref)
- Edetate disodium (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Incidence of adverse effects is dependent upon age of patient, dose, and route of administration.

>10%:

Nervous system: Excitement (children, adolescents: 2% to 20%), nervousness (4% to 15%), tremor (5% to 38%)

Respiratory: Bronchospasm (8% to 15%; exacerbation of underlying pulmonary disease), exacerbation of asthma (11% to 13%), pharyngitis (14%), rhinitis (5% to 16%), upper respiratory tract infection (5% to 21%)

1% to 10%:

Cardiovascular: Chest pain (<3%), edema (<3%), extrasystoles (<3%), hypertension (1% to 3%), tachycardia (1% to 7%) (table 1) Tachycardia

Table 1: **Albuterol: Adverse Reaction: Tachycardia**

Drug (Albuterol)	Placebo	Population	Dosage Form	Number of Patients
2%	N/A	Children	Oral syrup	
1%	N/A	Children, adolescents, and adults	Oral syrup	
7%	<1%	N/A	HFA inhalation aerosol	
2%	<1%	N/A	CFC 11/12 propelled inhaler	

Dermatologic: Diaphoresis (<3%), pallor (children: 1%), skin rash (<3%), urticaria ($\leq$ 2%)

Endocrine & metabolic: Diabetes mellitus (<3%)

Gastrointestinal: Anorexia (children: 1%), diarrhea (<3%), dyspepsia (1% to 2%), eructation (<3%), flatulence (<3%), gastroenteritis (3%), glossitis (<3%), increased appetite (children, adolescents: 3%), nausea (2% to 10%), unpleasant taste (inhalation site: 4%), viral gastroenteritis (1% to 3%), vomiting (3% to 7%), xerostomia (<3%)

Genitourinary: Urinary tract infection ($\leq$ 3%)

Hematologic & oncologic: Lymphadenopathy (3%)

Hypersensitivity: Hypersensitivity reaction (3% to 6%)

Infection: Infection (<3%; skin/appendage: $\leq$ 2%)

Local: Application-site reaction (HFA inhaler: 6%)

Nervous system: Anxiety (<3%), ataxia (<3%), depression (<3%), dizziness (<7%), drowsiness (<3%), emotional lability (1%), fatigue (1%), headache (3% to 7%), hyperactive behavior (children, adolescents: 2%), insomnia (1% to 3%), malaise (2%), migraine ($\leq$ 2%), pain (2%), rigors (<3%), shakiness (children, adolescents: 9%), vertigo, voice disorder (<3%)

Neuromuscular & skeletal: Back pain (2% to 4%), hyperkinetic muscle activity ($\leq$ 4%), lower limb cramp (<3%), musculoskeletal pain (3% to 5%)

Ophthalmic: Conjunctivitis (children: 1%)

Otic: Ear disease (<3%), otalgia (<3%), otitis media ($\leq$ 4%), tinnitus (<3%)

Respiratory: Bronchitis ($\geq$ 2%), cold symptoms (3%), cough ($\geq$ 3%), dyspnea (<3%), epistaxis (children, adolescents: 1%), flu-like symptoms (3%), increased bronchial secretions (2%), laryngitis (<3%), nasal congestion (1%), nasopharyngitis ($\geq$ 5%; children: 2%), oropharyngeal edema (<3%), oropharyngeal pain ($\geq$ 5%; children: 2%), pulmonary disease (<3%), respiratory system disorder (6%), sinus headache (1%), sinusitis ($\geq$ 5%), throat irritation (10%), upper respiratory tract inflammation (5%), viral upper respiratory tract infection (7%), wheezing (1% to 2%)

Miscellaneous: Accidental injury (<3%), fever (5% to 6%)

<1%:

Cardiovascular: Depression of ST segment on ECG

Gastrointestinal: Epigastric pain, stomach pain (children, adolescents)

Nervous system: Asthenia, irritability, sleep disturbance

Neuromuscular & skeletal: Muscle spasm (children, adolescents)

Ophthalmic: Mydriasis (children, adolescents)

Postmarketing:

Cardiovascular: Angina pectoris, atrial fibrillation, atrial tachycardia (Tandeter 2010), cardiac arrhythmia (Sears 2002), cardiomyopathy (takotsubo cardiomyopathy) (Khwaji 2016), hypotension (Ref), peripheral vasodilation, supraventricular tachycardia (Say 2014)

Dermatologic: Stevens-Johnson syndrome (Maggini 2015)

Endocrine & metabolic: Exacerbation of diabetes mellitus, hyperglycemia, hypokalemia (Udezue 1995), ketoacidosis, lactic acidosis (Ref), metabolic acidosis

Gastrointestinal: Dysgeusia, gag reflex, tongue ulcer

Hypersensitivity: Anaphylaxis, angioedema

Nervous system: Psychosis (Martin 1995; Whitehouse 1989)

Respiratory: Hoarseness, oropharyngeal irritation, paradoxical bronchospasm (Schissler 2018)

Contraindications

Hypersensitivity to albuterol or any component of the formulation; severe hypersensitivity to milk proteins (dry powder inhalers).

Canadian labeling: Additional contraindications (not in US labeling):

Injection: Ventolin: Hypersensitivity to albuterol or any component of the formulation; tachyarrhythmias; tocolytic use in patients at risk of premature labor or threatened abortion.

Inhalation: Hypersensitivity to albuterol or any component of the formulation; tocolytic use in patients at risk of premature labor or threatened abortion.

Warnings/Precautions

Disease-related concerns:

- Cardiovascular disease: Use with caution in patients with cardiovascular disease (arrhythmia, coronary insufficiency, hypertension, heart failure). In a scientific statement from the American Heart Association, albuterol has been determined to be an agent that may either cause direct myocardial toxicity or exacerbate underlying myocardial dysfunction (magnitude: moderate to major) (AHA [Page 2016]).
- Diabetes: Use with caution in patients with diabetes mellitus; beta-2 agonists may increase serum glucose and aggravate preexisting diabetes and ketoacidosis.
- Glaucoma: Use with caution in patients with glaucoma; may elevate intraocular pressure.
- Hyperthyroidism: Use with caution in hyperthyroidism; may stimulate thyroid activity.
- Hypokalemia: Use with caution in patients with hypokalemia; beta-2 agonists may decrease serum potassium.
- Renal impairment: Use with caution in patients with renal impairment.
- Seizures: Use with caution in patients with seizure disorders; beta-agonists may result in CNS stimulation/excitation.

Special populations:

- Pediatric: Oral inhalation: Use spacer for children <5 years of age and consider adding a face mask for infants and children <4 years of age.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol derivative with caution in neonates. See manufacturer's labeling.
- Lactose: Dry powder inhalers may contain lactose; hypersensitivity reactions (eg, anaphylaxis, angioedema, pruritus, and rash) have been reported in patients with milk protein allergy.

Other warnings/precautions:

- Appropriate use: Do not exceed recommended dose; serious adverse events, including fatalities, have been associated with excessive use of inhaled sympathomimetics.
- Patient information: Patients must be instructed to seek medical attention in cases where acute symptoms are not relieved or a previous level of response is diminished. The need to increase frequency of use may indicate deterioration of asthma, and treatment must not be delayed. A spacer device or valved holding chamber is recommended when using a metered-dose inhaler.

Dosage Forms Considerations

In the United States, the metered-dose inhaler (MDI) and dry powder inhaler (DPI) formulations provide 90 mcg/actuation (from the mouthpiece). In Canada, the MDI products are labeled as providing 100 mcg/actuation (per the valve depression [ie, from the valve]) and the DPI, Ventolin Diskus [Canadian product], provides 200 mcg/actuation.

ProAir Digihaler 0.65 g inhaler, ProAir HFA 8.5 g canisters, ProAir RespiClick 0.65 g inhaler, Proventil HFA 6.7 g canisters, and Ventolin HFA 18 g canisters contain 200 inhalations. Ventolin HFA 8 g canisters contain 60 inhalations. The ProAir Digihaler is a digital inhaler which detects, records, and stores inhaler events, including peak inspiratory flow rate, and transmits data to a mobile application for inhaler use information.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Aerosol Powder Breath Activated, Inhalation:

ProAir Digihaler: 108 (90 Base) MCG/ACT (1 ea [DSC]) [contains lactose monohydrate]

ProAir RespiClick: 90 mcg/actuation (1 ea) [contains milk protein]

Aerosol Solution, Inhalation:

ProAir HFA: 90 mcg/actuation (8.5 g [DSC])

Proventil HFA: 90 mcg/actuation (6.7 g [DSC])

Proventil HFA: 90 mcg/actuation (6.7 g [DSC]) [cfc free]

Ventolin HFA: 90 mcg/actuation (8 g, 18 g)

Generic: 90 mcg/actuation (6.7 g, 8.5 g, 18 g)

Nebulization Solution, Inhalation:

Generic: 0.63 mg/3 mL (3 mL); 1.25 mg/3 mL (3 mL); 0.083% [2.5 mg/3 mL] (3 mL); 0.5% [2.5 mg/0.5 mL] (30 ea, 20 mL [DSC])

Nebulization Solution, Inhalation [preservative free]:

Generic: 0.63 mg/3 mL (3 mL [DSC]); 1.25 mg/3 mL (3 mL); 0.083% [2.5 mg/3 mL] (3 mL); 0.5% [2.5 mg/0.5 mL] (1 ea, 30 ea)

Syrup, Oral:

Generic: 2 mg/5 mL (5 mL, 120 mL, 473 mL); 8 mg/20 mL (20 mL)

Tablet, Oral:

Generic: 2 mg, 4 mg

Generic Equivalent Available: US

May be product dependent

Pricing: US

Aerosol powder (ProAir RespiClick Inhalation)

108 (90 Base) mcg/ACT (per each): \$84.30

Aerosol solution (Albuterol Sulfate HFA Inhalation)

108 (90 Base) mcg/ACT (per gram): \$8.71 - \$9.31

Aerosol solution (Ventolin HFA Inhalation)

108 (90 Base) mcg/ACT (per gram): \$3.41

Nebulization (Albuterol Sulfate Inhalation)

0.63 mg/3 mL (per mL): \$0.28 - \$2.00

1.25 mg/3 mL (per mL): \$0.28 - \$2.33

(2.5 MG/3ML) 0.083% (per mL): \$0.12 - \$1.20

(5 MG/ML) 0.5% (per each): \$3.83 - \$5.00

Syrup (Albuterol Sulfate Oral)

2 mg/5 mL (per mL): \$22.23

Tablets (Albuterol Sulfate Oral)

2 mg (per each): \$5.87 - \$5.88

4 mg (per each): \$5.87 - \$5.88

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Aerosol Powder Breath Activated, Inhalation:

Ventolin Diskus: 200 mcg/actuation (1 ea) [contains lactose, milk protein]

Aerosol Solution, Inhalation, as sulfate:

Generic: 100 mcg/actuation (1 ea)

Aerosol Solution, Inhalation, as sulfate [strength expressed as base]:

Airomir: 100 mcg/actuation (6.7 g)

Ventolin HFA: 100 mcg/actuation (17 g)

Generic: 100 mcg/actuation (1 ea)

Nebulization Solution, Inhalation:

Ventolin: 0.5% [2.5 mg/0.5 mL] (10 mL)

Ventolin PF: (5 MG/2.5ML) 0.2% ([DSC]); (2.5 MG/2.5ML) 0.1% ([DSC])

Generic: (1.25 MG/2.5ML) 0.05% (2.5 mL); (2.5 MG/2.5ML) 0.1% (2.5 mL); (5 MG/2.5ML) 0.2% (2.5 mL); 0.5% [2.5 mg/0.5 mL] (10 mL)

Solution, Intravenous:

Ventolin: 1 mg/mL (5 mL)

Administration: Adult

IV: Infusion solution [Canadian product]: Do not inject undiluted. Reduce concentration by at least 50% before infusing. Administer as a continuous infusion via infusion pump.

Oral inhalation:

Metered-dose inhalers: Shake well before use; prime prior to first use, and whenever inhaler has not been used for >2 weeks or when it has been dropped, by releasing 3 to 4 test sprays into the air (away from face). Airomir [Canadian product] labeling recommends releasing a minimum of 4 test sprays when priming. HFA inhalers should be cleaned with warm water at least once per week; allow to air dry completely prior to use. A spacer device or valved holding chamber is recommended for use with metered-dose inhalers.

Dry powder inhalers:

ProAir Digihaler, ProAir RespiClick: Inhaler device is breath-actuated; does not require priming. Do not use with spacer or volume holding chamber. Keep inhaler clean and dry by wiping with dry cloth or tissue as needed; do not wash or put any part of inhaler in water.

Ventolin Diskus [Canadian product]: For oral inhalation route only. To activate Diskus, patient should slide lever using the thumb grip away from them as far as it will go (click should be heard). Before inhaling the dose, breathe out fully; do not exhale into the Diskus device. Bring mouthpiece to lips and inhale steadily and deeply through the Diskus; hold breath for about 10 seconds or for as long as comfortable and exhale slowly. To close Diskus slide thumb grip back as far as it will go towards its original position. To prevent a wasted dose, the lever should not be manipulated until administration of next dose. Diskus counts down from 60 to 1 and when 5 doses remain the numbers appear in red. Diskus should be kept dry.

Nebulization solution: Concentrated solutions ($\geq 0.5\%$) should be diluted prior to use; adjust nebulizer flow to deliver dosage over 5 to 15 minutes; avoid contact of the dropper tip (multidose bottle) with any surface, including the nebulizer reservoir and associated ventilator equipment. Blow-by administration is not recommended; use a mask device if patient unable to hold mouthpiece in mouth for administration. Compatibility with other medications (eg, budesonide, fluticasone, ipratropium) in nebulizer has been reported (Ref); also refer to institution-specific policies.

Oral (syrup, tablets): May administer without regard to meals.

Preparation for Administration: Adult

Nebulization solution: 0.5% solution: Dilute 0.25 mL (1.25 mg dose) or 0.5 mL (2.5 mg) of solution to a total of 3 mL with NS; also compatible with cromolyn or ipratropium nebulizer solutions.

IV [Canadian product]: Dilute 5 mg in 500 mL of NS, D5W, or D5NS for a concentration of 10 mcg/mL. Use within 24 hours.

Administration: Pediatric

Oral inhalation: In infants and children <4 years of age, a face mask with either the metered-dose inhaler or nebulizer is recommended (Ref).

Metered-dose inhaler: Shake well before use; prime the inhaler prior to first use and whenever it has not been used for >2 weeks or when it has been dropped by releasing 3 to 4 test sprays into the air away from the face. Airomir [Canadian product] labeling recommends releasing a minimum of 4 test sprays when priming. A valved holding chamber or spacer is recommended for use with an MDI in patients with poor technique and is the recommended medication delivery system in all patients ≤5 years of age, with the addition of a mask for patients <3 or 4 years of age or until proper technique is demonstrated (Ref). Remove cap from mouthpiece and shake inhaler. Exhale fully prior to bringing inhaler to mouth. Place inhaler or valved holding chamber mouthpiece in mouth, close lips around mouthpiece, and inhale slowly and deeply while pressing down on the canister with your finger. Remove device and hold breath for as long as possible, up to 10 seconds. Breathe out slowly. If prescribed dose is more than 1 inhalation, wait 1 minute between each inhalation and shake inhaler prior to each inhalation. HFA inhalers should be cleaned with warm water at least once per week; allow to air dry completely prior to use. Discard inhaler when the dose indicator window displays "0".

Dry powder inhaler:

ProAir Digihaler, ProAir RespiClick: Inhaler device is breath-actuated; does not require priming. Hold the inhaler upright and open cap. When cap is opened and click is heard, the dose is activated; do not open until ready for dose. Before inhaling the dose, breathe out fully; do not exhale into the mouthpiece. Place inhaler in mouth, close lips around mouthpiece, and inhale quickly and deeply. Hold breath for about 10 seconds or for as long as comfortable and exhale slowly. If additional inhalations are needed for a full dose, close cap and repeat the process used for the first inhalation. Do not use with spacer or volume holding chamber. Keep inhaler clean and dry by wiping with dry cloth or tissue as needed; do not wash or put any part of inhaler in water. The dose indicator tells how many doses are left. When the number 20 appears in red, only a few doses remain, and the pharmacy should be contacted for a refill. Discard inhaler 13 months after removal from foil pouch or when the dose counter reads "0" (whichever comes first).

Ventolin Diskus [Canadian product]: For oral inhalation route only. After opening the Diskus, hold in a level, horizontal position and activate by sliding the lever until it clicks. Before inhaling the dose, breathe out fully; do not exhale into the Diskus device; activate and use only in a level, horizontal position. Bring mouthpiece to lips and inhale quickly and deeply through the Diskus; hold breath for about 10 seconds or for as long as comfortable and exhale slowly. Do not use with a spacer device or wash mouthpiece; Diskus should be kept dry. The dose indicator tells how many doses are left. When the numbers 5 to 0 appear in red, only a few doses remain, and the pharmacy should be contacted for a refill.

Nebulization: Concentrated solutions (≥0.5%) should be diluted prior to use; avoid contact of the dropper tip (multidose bottle) with any surface, including the nebulizer reservoir and associated ventilator equipment. Blow-by administration is not recommended; use a mask device if patient is unable to hold mouthpiece in mouth for administration. Compatibility with other medications (eg, budesonide, fluticasone, ipratropium) in nebulizer has been reported (Ref); also refer to institution-specific policies

Oral: Syrup: Administer with an accurate measuring device (calibrated oral syringe or measuring cup); do not use a household teaspoon or tablespoon to measure dose (overdosage may occur).

Preparation for Administration: Pediatric

Solution for nebulization: 0.5% solution: Dilute 0.25 mL (1.25 mg dose) or 0.5 mL (2.5 mg dose) of solution to a total of 3 mL with NS; also compatible with cromolyn or ipratropium nebulizer solutions.

Storage/Stability

Metered-dose inhalers (HFA aerosols): Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Do not store at temperature >120°F. Do not puncture. Do not use or store near heat or open flame.

Proventil HFA: Store with mouthpiece down.

Ventolin HFA: Discard when counter reads 000. Store with mouthpiece down.

Dry powder inhalers:

ProAir Digihaler, ProAir RespiClick: Store between 15°C and 25°C (59°F and 77°F). Avoid exposure to extreme heat, cold, or humidity. Discard 13 months after opening the foil pouch, or when the counter displays 0, whichever comes first.

Ventolin Diskus [Canadian product]: Store at ≤30°C (86°F). Keep in a dry place. Protect from frost and light. Diskus is nonrefillable and should be discarded after all doses have been administered.

Infusion solution [Canadian product]: Ventolin IV: Store at 15°C to 30°C (59°F to 86°F). Protect from light. After dilution, discard unused portion after 24 hours.

Nebulization solution: Store at 2°C to 25°C (36°F to 77°F). Do not use if solution changes color or becomes cloudy. Products packaged in foil should be used within 1 week (or according to the manufacturer's recommendations) if removed from foil pouch.

Syrup: Store at 20°C to 25°C (68°F to 77°F).

Tablet: Store at 20°C to 25°C (68°F to 77°F).

Use: Labeled Indications

Bronchospasm: Treatment or prevention of bronchospasm in patients with reversible obstructive airway disease (eg, asthma).

Note: Asthma (oral product): Although included as an FDA-approved use in the manufacturer's prescribing information for the treatment of asthma, oral short-acting beta agonists, including oral albuterol, are not recommended due to the slower onset of action and higher risk of adverse effects (GINA 2024; NAEPP 2007).

Exercise-induced bronchospasm: Prevention of exercise-induced bronchospasm.

Use: Off-Label: Adult

Hyperkalemia (inhalation)

Medication Safety Issues

Sound-alike/look-alike issues:

Albuterol may be confused with Albutein, atenolol

Proventil may be confused with Bentyl, PriLOSEC, Prinivil

Salbutamol may be confused with salmeterol

Ventolin may be confused with phentolamine, Benylin, Vantin

International issues:

Durasal [India] may be confused with Durezol brand name for difluprednate [US, Canada]

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Atomoxetine: May increase hypertensive effects of Sympathomimetics. Atomoxetine may increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Atomoxetine: May increase tachycardic effects of Beta2-Agonists. Atomoxetine may increase hypertensive effects of Beta2-Agonists. *Risk C: Monitor*

Atosiban: Beta2-Agonists may increase adverse/toxic effects of Atosiban. Specifically, there may be an increased risk for pulmonary edema and/or dyspnea. *Risk C: Monitor*

Beta-Blockers (Beta1 Selective): May decrease bronchodilatory effects of Beta2-Agonists. *Risk C: Monitor*

Beta-Blockers (Nonselective): May decrease bronchodilatory effects of Beta2-Agonists. Management: Patients with asthma or COPD receiving beta-2 agonists should not normally be treated with non-selective beta-blockers. If used concomitantly, monitor closely for diminished bronchodilatory effects of the beta2-agonist. *Risk D: Consider Therapy Modification*

Beta2-Agonists (Short-Acting): May increase adverse/toxic effects of Beta2-Agonists (Short-Acting). Management: Avoid coadministration of more than one short-acting beta2-agonist whenever possible. If combined, use cautiously, and monitor for increased toxicities, particularly cardiovascular effects (eg, tachycardia, arrhythmias). *Risk D: Consider Therapy Modification*

Bitter Orange: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Bornaprine: Sympathomimetics may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Cannabinoid-Containing Products: May increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Cocaine (Topical): May increase hypertensive effects of Sympathomimetics. Management: Consider alternatives to use of this combination when possible. Monitor closely for substantially increased blood pressure or heart rate and for any evidence of myocardial ischemia with concurrent use. *Risk D: Consider Therapy Modification*

Dihydralazine: Sympathomimetics may decrease therapeutic effects of Dihydralazine. *Risk C: Monitor*

Doxapram: Sympathomimetics may increase adverse/toxic effects of Doxapram. *Risk C: Monitor*

Doxofylline: Sympathomimetics may increase adverse/toxic effects of Doxofylline. *Risk C: Monitor*

Droxidopa: Sympathomimetics may increase hypertensive effects of Droxidopa. *Risk C: Monitor*

Esketamine (Injection): May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for elevated heart rate, hypertension, and arrhythmias may be increased. *Risk C: Monitor*

Guanethidine: May increase hypertensive effects of Sympathomimetics. Guanethidine may increase arrhythmogenic effects of Sympathomimetics. *Risk C: Monitor*

Haloperidol: QT-prolonging Agents (Indeterminate Risk - Caution) may increase QTc-prolonging effects of Haloperidol. *Risk C: Monitor*

Kratom: May increase adverse/toxic effects of Sympathomimetics. *Risk X: Avoid*

Linezolid: May increase hypertensive effects of Sympathomimetics. Management: Consider initial dose reductions of sympathomimetic agents, and closely monitor for enhanced blood pressure elevations, in patients receiving linezolid. *Risk D: Consider Therapy Modification*

Loop Diuretics: Beta2-Agonists may increase hypokalemic effects of Loop Diuretics. *Risk C: Monitor*

Loxapine: Agents to Treat Airway Disease may increase adverse/toxic effects of Loxapine. More specifically, the use of Agents to Treat Airway Disease is likely a marker of patients who are likely at a greater risk for experiencing significant bronchospasm from use of inhaled loxapine. Management: This is specific to the Adasuve brand of loxapine, which is an inhaled formulation. This does not apply to non-inhaled formulations of loxapine. *Risk X: Avoid*

Methacholine: Beta2-Agonists (Short-Acting) may decrease therapeutic effects of Methacholine. Management: Hold short-acting beta₂ agonists for 6 hours before methacholine use. *Risk D: Consider Therapy Modification*

Monoamine Oxidase Inhibitors: May increase adverse/toxic effects of Beta2-Agonists. *Risk C: Monitor*

QT-prolonging Agents (Highest Risk): QT-prolonging Agents (Indeterminate Risk - Caution) may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Solriamfetol: Sympathomimetics may increase hypertensive effects of Solriamfetol. Sympathomimetics may increase tachycardic effects of Solriamfetol. *Risk C: Monitor*

Sympathomimetics: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Tedizolid: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for increased blood pressure and heart rate may be increased. *Risk C: Monitor*

Theophylline Derivatives: May increase hypokalemic effects of Beta2-Agonists. Beta2-Agonists may increase adverse/toxic effects of Theophylline Derivatives. Specifically, sympathomimetic effects may be increased. *Risk C: Monitor*

Thiazide and Thiazide-Like Diuretics: Beta2-Agonists may increase hypokalemic effects of Thiazide and Thiazide-Like Diuretics. *Risk C: Monitor*

Thyroid Products: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk of hypertension, tachycardia, and coronary insufficiency may be increased in patients with coronary artery disease. *Risk C: Monitor*

Tricyclic Antidepressants: May increase adverse/toxic effects of Beta2-Agonists. *Risk C: Monitor*

Pregnancy Considerations

Albuterol crosses the placenta (Boulton 1997).

Maternal use of beta-2 agonists is not associated with an increased risk of fetal malformations (GINA 2024). Systemic use may be associated with hypoglycemia and tachycardia in the mother and fetus (ERS/TSANZ [Middleton 2020]). Albuterol may affect uterine contractility by interacting with beta-2 adrenoceptors in the uterus.

Uncontrolled asthma is associated with adverse events on pregnancy (increased risk of perinatal mortality, preeclampsia, preterm birth, low-birth-weight infants, cesarean delivery, and the development of gestational diabetes). Poorly controlled asthma or asthma exacerbations may have a greater fetal/maternal risk than what is associated with appropriately used asthma medications. Maternal treatment improves pregnancy outcomes by reducing the risk of some adverse events (eg, preterm birth and gestational diabetes) (ERS/TSANZ [Middleton 2020]; GINA 2024).

Albuterol is the preferred short-acting beta-2 agonist (SABA) for the management of asthma during pregnancy (ERS/TSANZ [Middleton 2020]). Pregnant patients should be treated with a SABA for acute asthma exacerbations (GINA 2024). Maternal asthma symptoms should be monitored every 4 to 6 weeks during pregnancy (ERS/TSANZ [Middleton 2020]; GINA 2024). If high doses of albuterol are required within 48 hours of delivery, monitoring of glucose concentrations in the newborn for 24 hours is recommended, especially in preterm infants (GINA 2024).

Due to effects on uterine contractility, albuterol has been studied as a tocolytic to prevent preterm labor (not an approved use in the United States or Canada). Maternal side effects associated with this use include cardiac arrhythmias, metabolic changes (hypokalemia, hyperglycemia), myocardial

ischemia, and pulmonary edema; fetal tachycardia may also occur. Due to the increased risk of adverse events and the availability of safer options, albuterol is not recommended for use as a tocolytic (ACOG 2016; Besinger 1990; Chan 2006; NICE 2015; Sentilhes 2017; WHO 2015).

Breastfeeding Considerations

It is not known if albuterol is present in breast milk.

The amount of albuterol available systemically following inhalation is significantly less in comparison to oral doses. According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother.

Use of albuterol is generally considered compatible with breastfeeding when used in usual doses (ERS/TSANZ [Middleton 2020]; WHO 2002).

Monitoring Parameters

FEV₁, peak flow, and/or other pulmonary function tests; blood pressure, heart rate; CNS stimulation; serum glucose, serum potassium, serum creatinine; asthma symptoms; arterial or capillary blood gases (if patients condition warrants); lactate (IV formulation) (BTS 2019); continuous ECG monitoring (intravenous formulation).

Mechanism of Action

Relaxes bronchial smooth muscle by action on beta₂-receptors with little effect on heart rate.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

Nebulization solution: ≤5 minutes.

Oral inhalation: DPI: 5.7 minutes (median); MDI: 5.4 to 8.2 minutes (median).

Oral (syrup, tablet): ≤30 minutes.

Duration of action:

Nebulization solution: 3 to 6 hours.

Oral inhalation: DPI: ~2 hours (median); MDI: ~4 to 6 hours.

Oral (syrup, tablet): 6 to 8 hours.

Protein binding: 10%.

Metabolism: Hepatic to an inactive sulfate.

Half-life elimination:

Oral inhalation: 3.8 to ~5 hours.

Oral (syrup, tablet): 5 to 6 hours.

Time to peak, serum:

Nebulization solution: 30 minutes.

Oral inhalation: DPI: 30 minutes; MDI: 25 minutes (mean).

Oral (syrup, tablet): ≤2 hours.

Time to peak, FEV₁:

Nebulization solution: ~1 to 2 hours.

Oral inhalation: DPI: Within 30 minutes; MDI: 47 minutes.

Oral (syrup, tablet): 2 to 3 hours.

Excretion: Urine (80% to 100% [oral inhalation]; 76% over 3 days [60% metabolite; oral]); Feces (<20% [oral inhalation]; 4% [oral]).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: There was a 67% decline in albuterol clearance in patients with CrCl 7 to 53 mL/minute.

Patient Counseling Points

(For additional information see "Albuterol (salbutamol): Patient drug information")

What is this drug used for?

- It is used to open the airways in lung diseases where spasm may cause breathing problems.
- It is used to prevent exercise-induced breathing problems.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Nausea
- Vomiting
- Tremor
- Sore throat
- Back pain
- Aches
- Pains
- Runny nose

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Low potassium like muscle pain or weakness, muscle cramps, or an abnormal heartbeat
- Lung problems like shortness of breath or other trouble breathing, cough that is new or worse
- Chest pain
- Abnormal heartbeat
- Fast heartbeat
- Severe anxiety
- Severe headache
- Severe dizziness
- Passing out
- Vision changes
- Stevens-Johnson syndrome/toxic epidermal necrolysis like red, swollen, blistered, or peeling skin (with or without fever); red or irritated eyes; or sores in mouth, throat, nose, or eyes
- Painful urination
- Trouble urinating

- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Airomir | Asmadil | Asmanore | Asthalin | Asthavent | Asthavent dp | Butalin | Butovent | Disbron | Farcolin | Neoventil | Salamol | Vental | Ventamol | Ventol | Ventolin;
- (AR) Argentina: Aerolep | Aircosalm | Aircosalm hfa | Airsalbu | Airsalbu hfa | Amocasin | Asmatol | Asmatol hfa | Butamol | Cercini | Duopack | Labsavent | Microterol | Nebutrax | Niblet | Respirat | Salbucass | Salbuden | Salbuden con aerocamara df | Salbunib | Salbutamol | Salbutamol denver farma | Salbutamol fabra | Salbutamol hfa 3m | Salbutamol kilab | Salbutamol lacefa | Salbutamol lafedar | Salbutamol Pharma | Salbutamol richet | Salbutol | Salbutral | Salbutral c/aeromed | Salmobutal | Ventolin | Yontal | Zoom;
- (AT) Austria: Salbutamol sandoz | Sultanol;
- (AU) Australia: Airomir | Apo salbutamol | Asmol | Epaq | Respax | Respolin | Ventolin | Zempreon;
- (BD) Bangladesh: Absol | Actolin | Alvolex | Asmalin hfa | Asmatol | Asmolex | Asnil | Asthmarol | Asthmolin | Asul | Azmasol | Bitol | Broad | Brodil | Brolax | Broncoail | Broncolin | Bronkolax | Butalin | Butamol | Butol | D-Butamol | Decabutamol | Dilatol | G-salbutamol | Jp salbo | Koptolin | Proventa | Pulmocare | Pulmolin | Respolin | Reven | Salbo | Salbu | Salbunil | Salbusol | Salbut | Salbutal | Salbutamol | Salmol | Salmolin | Saltide | Sulbutamol | Sultolin | Sultolin hfa | Tolin | Venol | Ventisal | Ventol | Ventolex | Ventolin | Ventosol | Windel;
- (BE) Belgium: Airomir | Docsalbuta | Ecosal Easi Breath | Salbutamol sandoz | Ventolin;
- (BF) Burkina Faso: Asthalin | Buto Asma | Salbutus | Tamuxxair | Ventoline | Ventoline rotacaps;
- (BG) Bulgaria: Buto-Asma | Ecosal | Salamol | Salbutamol | Ventolin;
- (BR) Brazil: Acobelin | Aero ifal | Aero-ped | Aeroclean | Aerodini | Aerofrin | Aerogold | Aerogreen | Aerojet | Aerolin | Aeromed | Aerotamol | Aerotrat | Albulin | Asmaflux | Asmakil | Asmaliv | Broncofedrin | Broncomix | Bronconal | Butalab | Butovent | Dilamol | Lafepe salbutamol | Neutoss | Pulmoflux | Regrair | Salburin | Salbutalin | Salbutam | Salbutamax | Salrolin | Saltamol | Sarolin | Sulfato de salbutamol | Teoden | Tussiliv;
- (CH) Switzerland: Ecovent | Salamol | Ventodisk | Ventolin | Ventolin zuckerfrei mit saccharin | Volmax;
- (CI) Côte d'Ivoire: Aerol | Asthalin | Broncoron | Salbutus | Ventoline rotacaps;
- (CL) Chile: Aero-sal | Aerolin | Astin | Broncoterol | Bropil | Butotal | Fesema | Respirat | Respolin | Sacrusyt | Salbusev | Salbutamol | Salbutral | Salbutral hfa | Sinasmal;
- (CN) China: Ai sha xi le | Cyclocaps | Da fen ke chuang | Hui bai shi | Pin chuan | Salbutamol | Sha pu er | Shuan shu ning | Ventolin | Ventolin Nebules | Volmax | Wan tuo lin;
- (CO) Colombia: Airmax | Asmepoc | Asthalin Hfa | Bajapres | Bentusol | Benylan | Betadren |

- Bonair | Bronter | Brotamol | Bugonol | Butotal | Ciplabutol | Deltabutol | Friotan | Neumoterol | Parasma | Rexpirex | Sacrusyt | Salbumed | Salburex | Salbutamol | Salbutamol MK | Salbutamol mk | Salbutan | Servitamol | Solbucort | Ventilan;
- (CR) Costa Rica: Broncomat | Buto air | S.b.t. 4 | Salbudil | Salbutamol | Salbutamol base | Salbutin | Salbutral | Salden | Sultolin | Ventolin;
 - (CZ) Czech Republic: Asmaven | Asthalin | Buventol | Ecosal | Salbumol | Salbutamol | Ventolin;
 - (DE) Germany: Aerolin | Aerolind | Albuterol hikma | Apsomol | Apsomol n | Broncho | Bronchospray | Loftan | Paediamol | Salbu | Salbu-fatol | Salbuhexal | Salbuhexal N | Salbupur d | Salbupur e | Salbusandoz | Salbutamol | Salbutamol 1 A Pharma | Salbutamol al | Salbutamol aldo union | Salbutamol CT | Salbutamol sandoz | Salbutamol stada | Salbutamol Trom | Salbutamol-ratiopharm n | Salvent | Sultanol | Sultanol forte | Volmac;
 - (DO) Dominican Republic: Aerolin | Airplen | Albugenol | Alunic | Amocasin | Asthalin | Astumol | Broncomat | Broncosamol | Buto air | Buto Asma | Butovent | Dilabron | Elbutamol | Frabutamol | Lascobutamol | Nebusol | Respiret | Salbucyma | Salbufar | Salbulin | Salbumox | Salbutamol | Salbutamol Feltrex | Salbutan | Salbutol | Salbutral | Salcrymol | Salventil | Ventolin;
 - (EC) Ecuador: Airmax | Airplen | Asmavent sf | Asthalin | Bemim | Bonair | Broncoterol | Butotal | Ciplabutol | Ecutamol | Farmarest | Kro salburol | Krosalburol | Nebsal | Nebulasma | Sacrusyt | Salbutamol | Salbutamol Ecar | Salbutamol mk | Salbutamol MK | Salbutero | Salbutral | Salden | Ventolin;
 - (EE) Estonia: Buventol | Ecosal | Ecosal easi-breath | Salbutamol ratiopharm | Salbutamol-dynapharm | Salbuvent | Ventolin | Volmax;
 - (EG) Egypt: Aerolin | Albuterol | Asthmarelief | Broncholin s | Bronchoterol | Bronchovent | Butalin | Farcolin | Mepacovent | Octovent | Salbovent | Salbuline | Salbutamol | Vental | Ventene | Ventolin | Viasalmol;
 - (ES) Spain: Aldobronquial | Bronsal | Buto air | Buto Asma | Dipulmin | Emican | Salbutamol aldo union | Salbutamol sandoz | Ventiloboi | Ventoaldo | Ventolin;
 - (ET) Ethiopia: Albuterol | Lebasma | Salbutamol | Salbutamol sulphate | Ventolin;
 - (FI) Finland: Airomir | Buventol | Salbumol | Ventoline;
 - (FR) France: Airomir | Salbumol | Spreor | Ventexxair | Ventoline;
 - (GB) United Kingdom: Airomir | Airsalb | Asmasal | Asmavent | Cobutolin | Libetist | Salamol | Salapin | Salbulin | Salbutamol | Salbutamol Berk | Salbutamol Ivax | Salbutamol sandoz | Ventolin | Ventolin evohaler;
 - (GR) Greece: Aerolin | Asthmotrate | Normobron | Normobron pulvinal | Salbunova | Salbutamol | Salomol | Ventolin;
 - (GT) Guatemala: Aeromax | Airmax | Asmatol baby bon | Salbutamol | Salbutamol mk | Ventolin;
 - (HK) Hong Kong: Airomir | Apo salvent | Asmalin | Asmovent | Asthalin Hfa | Axcel Salbutamol | Azmacon | Azmasol | Buto Asma | Cybutol | Eurotolin | Medolin | Respolin | Respreve | Salamol | Salbutamol | Salmol | Saltalin | Salven | Solia | Syntalin | Synvent hfa 100 | Uni-Butamol | Vantin | Ventamol | Venteze | Ventolin | Zenmolin;
 - (HR) Croatia: Salbutamol Pliva | Ventolin;
 - (HU) Hungary: Ac-butamol | Ecosal | Huma-salmol | Salbutamol | Ventolin | Volmax;
 - (ID) Indonesia: Asmacare | Asmacel | Astharol | Azmacon | Brondisal | Cybutol | Dilatamol | Dipsamol | Fartolin | Glisend | Grafalin | Lasal | Librentin | Pritasma | Respiron | Respolin | Salbron | Salbulin | Salbutamol | Salbuven | Saltam | Suprasma | Venterol | Ventolin;
 - (IE) Ireland: Airomir | Gerivent | Salamol | Salbul | Salbutamol | Salomol | Ventamol | Ventolin;
 - (IL) Israel: Apovent | Ventolin | Volmax;
 - (IN) India: Aerotaz | Alsai | Asmanil | Asthalin | Asthalin Hfa | Asthalin sa | Astinol | Bemol | Broncocet | Bronko | Bronkonat | Brosol | Derihaler | Durasal | Frespire | Lebasma | Pertulyn | Rheolin | Salbair | Salbetol | Salbid | Salbusun | Salbutamol | Salbutas | Salmaplone | Somavent | Sos | Vent | Ventamol | Ventil | Ventisol | Ventorlin | Ventryl;
 - (IQ) Iraq: Buetarin d | Butadil j | Butadin | Butamol | Nutadeen | Salbutamol | Salbutin | Salutin | Venolin mdi | Ventalen | Ventodad | Ventomel | Ventosaf;
 - (IT) Italy: Aerotec | Broncovaleas | Salbufax | Salbutamol Sandoz | Ventolin;
 - (JO) Jordan: Aerolin | Asmadil | Asmanor | Asthalin | Butalin | Buto-Asma | Butovent | Medolin | Salbutis | Vental | Ventol | Ventolin;
 - (JP) Japan: Airomir | Asmidon | Asu tajisu | Benareal | Renapirin | Salbutamol | Saltan | Sultanol

- | Tivoline | Venetlin;
- (KE) Kenya: Asbutol | Asmeez | Astalin | Asthadin | Asthalin | Axiomol | Azmasol | Bonair | Brodil hfa | Bronchomol | Butin | Derihaler | Lastmol | Liscoril | Medisalan | Sabulin | Salbutocort | Salbumol | Salbut | Salbutamol | Salfil | Statasmin | Sultolin hfa | Tabusol | Tavent | Venteze | Ventil | Ventolin | Ventomol | Ventosal;
- (KR) Korea, Republic of: Respolin | Salbutamol | Salbutamol bkw | Salbutol | Ventolin | Ventolin respirator;
- (KW) Kuwait: Butalin | Neoventil | Salamol | Salbucare | Salbutamed | Vental | Ventol | Ventolin | Ventomax;
- (LB) Lebanon: Alopriol | Asmalat | Butalin | Buto Asma | Butovent | Neoventil | Salbutamol | Ventolin | Viasalmol | Volmax;
- (LT) Lithuania: Asmaven | Asthalin | Astmasol | Ecosal | Salamol | Salbutamol | Salbutamol eipico | Salbutamol inteli | Salbuvent | Ventodisk | Ventolin | Volmax;
- (LU) Luxembourg: Airomir | Ventolin;
- (LV) Latvia: Asmaven | Asthalin | Astmasol | Salamol | Salbutamol | Salbutamol inteli | Salbutamol maxpharma | Salbuvent | Sultanol | Ventolin | Volmax;
- (MA) Morocco: Aerol | Bulmol | Butamyl | Butovent | Dilator | Inaler | Servitamol | Ventoline;
- (MX) Mexico: Alisal | Aposalvent | Assal | Avedox fc | Avendox fc | Azyrol | Biorenyn | Bresaltec | Brusol | Capacit | Cobamol | Exafil | Farmarest | Inbumed | Iqfavent | Nebul | Quinfaval | Rodribrak | Sacrusyt | Salamol | Salbutalan | Salbutamol | Salbutamol boehringer | Salbutamol gi | Salbutamol gi arle | Salbutamol gi brul | Salbutamol gi merc | Salbutamol gi salu | Salbutamol protein | Salbutamol ultra | Salbutamol wyeth | Salbutamol zerboni | Salcomed | Sofamix | Tunxin | Unibron | Ventilan | Ventolin | Vetrivatox | Zibil;
- (MY) Malaysia: Airomir | Anasma | Asmaliv | Asmin | Asmol | Asthalin | Asthavent | Axcel Salbutamol | Azmasol | Beatolin | Biomol | Bonair | Breathnine | Brethmol | Broncolin | Butahale | Butavent | Butovent | Healol salbutamol | Medolin | Rescanil | Salamol | Salbutamol | Salmox | Salmol | Saltoline | Ventamol | Ventolin;
- (NG) Nigeria: Afrab salbutamol | Asmacof p | Asthalin | Axcel salbutamol | Azmasol | Bexilin | Bleornol | Bonair | Farmex salbutamol | Greenhope salbutamol | Kinzolin | Sabumalin | Salbutamol | Suretolin | Zotrex;
- (NL) Netherlands: Aerolin | Airomir | Salbutamol | Salbutamol A | Salbutamol actavis | Salbutamol CF | Salbutamol cyclocaps | Salbutamol pch | Salbutamol redihaler | Salbutamol sandoz | Salbutamol Teva | Ventolin | Ventolin es;
- (NO) Norway: Aerolin | Airomir | Salbutamol eipico | Ventilan | Ventolin | Ventoline;
- (NZ) New Zealand: Airomir | Asmol | Broncolin | Respigen | Respolin | Salair | Salamol | Salbutamol | Ventolin;
- (PA) Panama: Aerolin | Airmax | Asthalin | Asthalin Hfa | Broncomat | Bronquit | Butamol | Buto Asma | Farmarest | Respilar | Sacrusyt | Salbusev | Salbutamol | Salbutamol MK | Salbutral | Ventolin;
- (PE) Peru: Acteril | Aeroler | Aircosalm hfa | Airplen | Apo salvent | Asthalin | Broncobutol | Broncobutol plus nf | Bronquimax | Butahale | Butotal | Pectolin | Salbid | Salbubronc | Salbusev hfa | Salbutam | Salbutamo | Salbutamol | Salbutex | Salbutol | Salbutral | Salbutrin | Salbuvent | Servitamol | Sultadrin | Ventimax | Ventolin | Ventusol;
- (PH) Philippines: Activent | Airhexal | Airomir | Airlsal | Albulem | Almodex | Amoltex | Arbadol | Asbunyl | Asfrenon | Asmacaire | Asmafort | Asmalin | Asmamed | Asmar | Asmarex | Astagen | Asthagen | Asvimol | Axmaxolv | Azmasol | Beisol | Bioxal | Boie Salbutamol | Breezol | Broncaire | Bronchosar | Broncolin | Bronelid | Brytolin | Butadil | Butovent | Cletal | Derihaler | Efamed | Emplusal | Gloleen | Hivent | Librentin | Medic aid salbutamol | Meventil | Mgl-salbutamol | Nobutol | Primesal | Prox-s | Resdil | Resovent | Respolin | Sal | Salbuflo | Salbuhaler | Salbumed | Salbumin | Salbusol | Salbutamol | Salbutamol Am-Europharma | Salbutamol Myrex | Salbutamol Pacific | Salbuvar | Salmate | Saltal 4 | Salvex | Sedalin | Socamol | Solben | Theoryl | Turbolax | Venalax | Vencronyl | Vent | Ventar | Vento broncho | Ventolax | Ventolin | Ventomax | Ventosal | Ventress | VN2;
- (PK) Pakistan: Aerolin | Aerotec | Asmatol | Asthamol | Asthavent | Blissbronchial | Breathmore | Bronchilate | Broncolin | Bronkal | Broven | Butamin | Butamol | Butoasma | Butovent | Erobinol | Eulin | Faktolin | Inhalerlin | Inspirol | Radisal | Respilax | Respolin | Salbair | Salbax | Salbest |

- Salbo | Salbovent | Salbree dpi | Salbuhaler | Salmol | Venex | Ventiplus | Ventolin | Ventopharm | Ventosal | Wintol | Zaftolin;
- (PL) Poland: Albuterol | Aspulmo | Sabumalin | Salamol eb | Salbutamol | Salbuvent | Velaspir | Ventolin;
 - (PR) Puerto Rico: Albuterol | Albuterol Hfa | Albuterol sulfate | Albuterol sulfate hfa | Proventil | Proventil hfa | Ventolin | Ventolin hfa;
 - (PT) Portugal: Prodome | Salbutamol Generis | Salbutamol GP | Salbutamol sandoz | Ventilan;
 - (PY) Paraguay: Acteril | Acteril plus | Aircosalm hfa | Asmavent sf | Asmazol | Asthalin | Broncofil | Brondil plus | Butamol | Codetamol | Lung | Respiral | Respiran | Salbutamol caplin point | Salbutamol dutriec | Salbutamol empa | Salbutamol genfar | Salbutamol imedic | Salbutamol la sante | Salbutamol pasteur | Salbutamol v y t farma | Salbutamol vyt farma | Salbutol | Salbutolin | Salbutral | Sinasmasal | Sinasmasal sf | Tarison | Ventilex | Ventiplus | Ventolin | Zoom;
 - (QA) Qatar: Asthavent | Asthavent Respules | Breathnine | Butalin | Rescuwave | Salbo HFA | Salbolin | Tamol-Q | Ventol | Ventolin | Ventolin Diskus | Ventolin Evohaler | Ventolin Nebules | Ventolin Respirator | Volmax;
 - (RO) Romania: Farcolin | Salbutamol eipico | Ventolin;
 - (RU) Russian Federation: Astalin | Asthalin | Cybutol cyclocaps | Gen salbutamol | Novatron | Novatron neo | Salamol | Salamol easy breat | Salamol eco | Salamol eco easy breath | Salbutamol | Salbutamol aeronativ | Salbutamol air | Salbutamol av | Salbutamol mhfp | Salbutamol pharmstandart | Salbutamol Teva | Salbuvent | Salgim | Saltos | Ventocol | Ventodisk | Ventolin | Vertasort | Volmax;
 - (SA) Saudi Arabia: Airomir | Asmadil | Asmanore | Asthalin | Asthavent | Butalin | Butovent | Disbron | Farcolin | Neoventil | Salamol | Salbulase | Salbutamed | Ventolin | Ventolin cr | Vintec;
 - (SE) Sweden: Airomir | Sabufarm | Salbutamol Scand Pharm | Salbuvent | Ventoline | Ventoline evohaler;
 - (SG) Singapore: Apo salvent | Asmol | Azmasol | Butahale | Buto-Asma | Medolin | Sabutol | Salamol | Salbuair | Salbutamol | Salmol | Unisabmol | Venderol | Ventamol | Ventolin | Zenmolin;
 - (SI) Slovenia: Ventolin;
 - (SK) Slovakia: Ecosal | Salamol | Salbutamol | Ventodisk | Ventolin;
 - (SL) Sierra Leone: Chemolin;
 - (SR) Suriname: Apo salbutamol | Apo salvent | Asthalin | Azmasol | Buto Asma | Salbutamol | Salbutamol MK | Salmaplone | Sultolin | Sultolin hfa | Ventamol | Ventolin;
 - (TH) Thailand: Aeromol | Aerotamol | Antomol | Asmasal | Asmatol | Asmatum | Asmo | Asmol | Asthalin | Asthamol | Asthma | Asthmolin | Bocoma | Bonair | Bronchodyl | Bronchosol | Butamol | Buto-asma inhaler | Butovent | Cintamol | Kressvent | Medolin | N-tolin | Naso | Natolin | Respolin | Sabumol | Saburin | Salamol | Salbetol | Salbulin | Salbusian | Salbutac | Salbutamol | Salbuvent | Salda | Saldol | Salmodon | Salmol | Saltalin | Saltolin | Satalin | Servitamol | Solia | Starbumol | Tanalin | Venbumed | Venbumol | Venbutin | Venline | Venmed | Ventamed | Venterol | Ventolin | Violin;
 - (TN) Tunisia: Albutol | Asmadil | Butalin | Butovent | Salbutamol | Salbuvent | Ventmax | Ventoline;
 - (TR) Turkey: Asthavent | Brecur | Butovent | Salbulin | Salbutol | Salres | Vent o sal | Ventolin | Ventosal;
 - (TW) Taiwan: Airomir | Albuterol | Albutol | Astamol | Bentolin | Censolin | Cotran | Mozal | Nopant | Respolin | Sabuchan | Sabumol | Sabutal | Sadamol | Salamol | Salbumol | Salbutain | Salbutamo | Salbutamol | Salbutamol c.h. | Salbutan | Salbutol | Salbuvent | Salmol | Saltmol | Saltol | Saltolin | Salutol | Samol | Satamol | Sucotin | Sunling | Synvent hfa | Ventolin;
 - (UA) Ukraine: Aerolin | Asthalin | Brutamol | Nebutamol | Salamol | Salamol Easy Breath | Salamol eco | Salbutamol | Salbutamol neo | Salbuvent | Ventolin;
 - (UG) Uganda: Ago lin | Agobutol | Agolin | Asbutol | Asthalin | Astharen | Azmasol | Kam vent | Lebasma | Sabulin | Salbumol | Salbut | Salres | Ventolin | Ventomac;
 - (UY) Uruguay: Airomir | Asmatol | Butovent | Raudosal | Respiran | Salbulin | Salbutamol Winpharm | Salbutral | Tarison | Ventiplus | Ventolin;
 - (VE) Venezuela, Bolivarian Republic of: Aerodini | Albutan | Asthalin | Butahale | Butoas | Diadilat | Respolin hfa | Sacrusyt | Salbuden | Salbulis | Salbumed | Salburol | Salbutacare | Salbutamol | Salbutamol aldo union | Salbutan | Salbutral + aeromed | Salres | Sultolin | Ventolin;
 - (VN) Viet Nam: Hasalbu | Pro salbutamol | Salbuboston | Zensalbu;

- (ZA) South Africa: Airomir | Asthalin | Asthavent | Azmaler | Cybutol | Glenbute | Proventil | Saasth | Salbulin | Vari salbutamol | Venteze | Ventimax | Ventolin | Viavent;
- (ZM) Zambia: Asthalin | Asthamol | Lastmol | Lebasma | Mestil | Sabulin | Salbair | Salbamol | Salbukant | Salbuta | Salbutamol | Salfil | Salmol | Samaol | Samol | Statasmin | Ventamol | Venteze | Ventolin;
- (ZW) Zimbabwe: Asmanol | Asthalin | Salbutamol | Venteze | Ventolin

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